PFP: Compression that preserves future operations and their cost

Jeremy H. Carroll

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Code baseline: 59dde95a7860063c25d75db37f40af556e9f28ee

Abstract

Equal current answers need not permit equal future choices. For a finite deterministic transition table, we characterize an exact operation and cost quotient and construct the coarsest stable refinement of an initial report. We give counterexamples to output-only compression and to choosing the fewest operations as the cheapest solution. The finite result imports standard transition-system refinement; its WDP role is preserving the original future question during representation change.

Mechanism inventory

Mechanism	Evidence
Mathematical statement	Written proof under declared hypotheses
Finite example	Standard-library replay of submitted inputs and rejection controls
Model inference	Not executed
Lean verification	No new formalization claimed
Authorization	None supplied

1. The declared finite model

Let X be a finite state set, A a finite action alphabet, and q_0 an initial report. Each action a has a partial deterministic transition t_a and a positive rational cost c_a on its domain. Terminal task labels are included in the protected report. A quotient $q : X \to Z$ is exact if all action domains, costs, successor reports, and terminal labels descend and reflect the declared concrete operations.

2. Necessary and sufficient structural conditions

Theorem. Such an exact quotient exists on a supplied report q precisely when states in one fibre have the same terminal labels and, for each action, the same enablement, the same enabled cost, and the same successor fibre.

Proof. If quotient operations descend and reflect, evaluating one quotient state cannot depend on which representative was chosen. Each listed condition follows. Conversely, define enabled quotient actions, their costs, and their successors using any representative. The conditions make all these definitions independent of the representative. They give the commuting step maps and reflection of action enablement. Induction on a word then gives identical legal action words, protected reports, and accumulated costs for states in one fibre. Consequently the best finite resolution cost is also preserved, including the absence of any legal resolving word.

This theorem concerns a deterministic table. For adversarial outcomes, the required successor data are the complete declared outcome relations in the appropriate information-state arena; Appendix

S states that stronger model. Deterministic examples cannot certify a stochastic or hidden-state adapter.

3. Coarsest stable refinement

Start from q_0's partition, including terminal labels. Repeatedly split each block by the action signature: disabled, or the pair consisting of exact cost and successor block. Each proper split increases the number of blocks, so at most $|X| - 1$ proper splits occur. A stable partition satisfies the theorem's conditions. Any stable refinement of q_0 refines every iteration: prove this by induction using its equal costs and successor classes. Therefore the terminal partition is the coarsest stable refinement. This is a finite construction, not a physically supplied observation instrument.

4. Current-output and cost counterexamples

Two states a, b both report zero, and both allow `finish`, reaching the same terminal state. Its cost is one at a and three at b. The current output is preserved by identifying them, but operating cost is not. Disabling `finish` at b also destroys future capability. Conversely, a task asking only for the current zero has the same answer in both states despite these structural failures: accidental task-value preservation is weaker than operation-preserving exactness.

In a second table, `direct` reaches the terminal in one operation costing five. `next` and `finish` take two operations costing one each. The cheaper route costs two. Add a positive-cost self-loop: it cannot improve any finite resolving word because deleting its repeated-state segment gives a legal word of strictly smaller cost. Thus an optimum, when reachable, occurs among simple paths of at most $|X| - 1$ transitions in this deterministic model.

5. Replay and scope

The cost example declares states `r`, `s` and terminal `t`: `r` can go directly to `t` for 5, move to `s` for 1 and then finish for 1, or loop at `r` for 1. The checker replays the submitted route and its claimed rational price. A separate procedure enumerates simple paths in the original table. Positive cycles can be erased from a terminating deterministic route, so that finite enumeration establishes the minimum 2. A falsely claimed price, an expensive route presented as optimal and an unfinished route are rejected.

The reference computes the stable refinement, checks exact rational route costs, and preserves a positive self-loop as a rejection control for the mistaken minimal-operation rule. It makes no provider calls. The new theorem has a written proof and finite replay, with no new Lean claim. The open native Source-25 target still requires actual states, edits, guard receipts, costs, and a declared carrier bound; this abstract table does not supply those bindings or implement a physical observer.

The two-level teaching packet includes shared system rules, this typed user instance, a different worked example and a required candidate response schema. It keeps provider system/user roles separate. It is a saved example, not a schema-admission or model-run receipt; the local replay checks the mathematical or protocol obligations stated above.

Research extension: bounded context without a preservation claim

The AIDE2-inspired reference replaces full attempt history with the root, champion, two recent attempt summaries and references to retained evidence. The host refuses a projection that drops required references. Failure guidance is scoped and capped. These are bounded operational choices,

not a construction of the coarsest stable quotient defined above. The external paper reports prompt size reductions for its own benchmarks (p. 9); those values are not local results.

Consider two resumption records with the same displayed answer and payload digest. One retains the artifact reference, task and state binding; the other keeps only the digest. A later resume operation can be enabled on the first and refused on the second. A shorter summary that identifies them therefore violates future-operation preservation even if their current displayed answers agree. Keeping a textual reference is necessary for the campaign's custody contract, but is not sufficient to prove that a stochastic model will use the referenced content correctly. Equal token counts likewise imply no equality of future costs, action availability or successor states.

The campaign records context bytes, actual reported input and output tokens, completion, failures and cost. The no-compression ablation keeps full prior code within the same request ceiling; crossing that ceiling is a retained refusal. This compares bounded policies rather than granting one arm an unlimited context. A reduction in context size may accompany a loss of useful distinctions, and a correct answer on the measured panel does not discharge every future operation.

The existing finite deterministic theorem and its replay are reused unchanged. No new Lean result is claimed. Since the live campaign has not run, this revision reports no local compression factor, cost saving or preservation measurement.

The source and page references for this extension are collected in Paper 029 and its source-to-claim map. The author-review rendering does not replace a historical deposit or archive.

Sources and relation to the programme

This draft builds on the WDP manuscript, Paper 021's source-and-execution distinctions, and the corrected findings notices. Paper 022 supplies a research design whose live-model and offline-strategy claims require their own source-bound scrutiny. These drafts do not adopt its empirical tables as new measurements. The PFP collection guide provides the shared-system/typed-instance protocol. Genesis and MSS-N presentation stages do not license a new mathematical or physical carrier.